

Milestone 2 Implementation Report

Project: CMPG325-2026-066
Client: CLI-066 Kagiso Mechanical Engineering (Rustenburg)
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1. Scope

Milestone 2 converts the Milestone 1 draw-only topology into a working Cisco Packet Tracer implementation. The assigned technical challenge is IPv4 VLSM subnetting using `192.168.35.0/24`. The client change request is a new file/application server that is reachable only by authorised departments.

This report records the final device configuration, verification output, test results, screenshots, and troubleshooting history for the completed Milestone 2 implementation.

2. Addressing baseline

VLAN	Purpose	Subnet	Gateway
10	Admin / HR	192.168.35.32/28	192.168.35.33
20	Engineering Design	192.168.35.0/27	192.168.35.1
30	Production Workshop	192.168.35.48/28	192.168.35.49
40	Finance / Procurement	192.168.35.96/28	192.168.35.97
50	ICT Technical	192.168.35.64/28	192.168.35.65
99	Management / Servers	192.168.35.112/28	192.168.35.113
100	Guest WiFi	192.168.35.80/28	192.168.35.81
Core link	R-EDGE to DLS1 routed link	192.168.35.252/30	R-EDGE .253, DLS1 .254

Implementation correction: `192.168.35.112` is the VLAN 99 network address and must not be assigned to DLS1. DLS1 uses `192.168.35.113` as the VLAN 99 SVI/management address. SVR-DC uses `192.168.35.114`.

3. Per-device configuration summary

Device	Hostname	Key IP addresses	Services and features
Distribution switch	DLS1	VLAN 10 .33/28, VLAN 20 .1/27, VLAN 30 .49/28, VLAN 40 .97/28, VLAN 50 .65/28, VLAN 99 .113/28, VLAN 100 .81/28, routed link .254/30	Inter-VLAN routing, DHCP relay, OSPF Area 0, Rapid-PVST+ root primary, trunks, ACLs, port security
Core router	R-EDGE	LAN .253/30, WAN address obtained by DHCP where supported	OSPF Area 0, default route, NAT/PAT configuration
Admin access switch	ALS-ADM	VLAN 99 management .115/28	VLAN 10 access ports, Gig0/2 trunk to DLS1, PortFast, BPDU

			Guard, port security
Engineering access switch	ALS-ENG	VLAN 99 management .116/28	VLAN 20 and 50 access ports, redundant trunks on DLS1 Fa0/21 and Fa0/22, PortFast, BPDU Guard, port security
Production access switch	ALS-PRD	VLAN 99 management .117/28	VLAN 30 and 100 access ports, Fa0/23 trunk to DLS1, PortFast, BPDU Guard, port security
Finance access switch	ALS-FIN	VLAN 99 management .118/28	VLAN 40 access ports, Fa0/24 native VLAN 999 trunk to DLS1, PortFast, BPDU Guard, port security
Domain/DHCP/DNS server	SVR-DC	192.168.35.114/28, gateway .113	GUI-configured DHCP with seven scopes, GUI-configured DNS, kagiso.local record
File/application server	SVR-FP	192.168.35.119/28, gateway .113	GUI-configured HTTP service and file/application endpoint

4. Planned implementation

- Configure VLANs 10, 20, 30, 40, 50, 99, and 100 on DLS1 and access switches.
- Configure access ports and trunk allow-lists according to the Milestone 1 logical topology.
- Configure DLS1 SVIs, `ip routing`, and DHCP relay to `192.168.35.114`.
- Configure the R-EDGE to DLS1 routed `/30` link.
- Configure OSPF Area 0 between R-EDGE and DLS1.
- Configure the default route and PAT on R-EDGE where supported by the Packet Tracer ISP model.
- Configure DHCP scopes on SVR-DC using the VLSM ranges and exclusions.
- Configure the file/application server on VLAN 99 and restrict access with ACLs.
- Configure port security, PortFast, BPDU Guard, Rapid-PVST+, and the Engineering redundant-link failover demonstration.

5. Acceptance tests

ID	Test	Expected result	Evidence
VLSM-01	Verify VLAN 20 mask, network, gateway, and broadcast	192.168.35.0/27, gateway .1, broadcast .31	PASS - 26_dls1_vlsm_svis_and_helpers.png
VLSM-02	Verify all seven VLAN ranges do not overlap	All ranges remain inside 192.168.35.0/24	PASS - 26_dls1_vlsm_svis_and_helpers.png
VLSM-03	Obtain DHCP leases in representative client VLANs	Correct subnet, gateway, and DNS are received	PASS - 01_engineering_dhcp.png, 02_admin_dhcp.png, 08_guest_dhcp.png, 09_finance_dhcp.png

ROUTE-01	Verify OSPF neighbour and learned routes	R-EDGE and DLS1 form an Area 0 adjacency	PASS - 11_dls1_ospf.png, 12_dls1_route.png
CORE-01	Ping an authorised host across VLANs	ICMP succeeds where policy permits	PASS - 03_engineering_server_ping.png
SERVER-01	Access the file/application server from Engineering	Authorised access succeeds	PASS - 03_engineering_server_ping.png
SERVER-02	Access the file/application server from Production	Access is denied	PASS - 04_production_server_ping_denied.png
SEC-01	Attempt Finance/Production access to the server	ACL blocks the traffic	PASS - 04_production_server_ping_denied.png, 13_dls1_acls.png
SEC-02	Verify management ACL policy	Non-authorised management traffic is denied	PASS - 13_dls1_acls.png
STP-01	Verify the Engineering redundant uplink state	STP identifies the root and forwarding/alternate paths	PASS - 15_stp_before_failover_part1.png, 16_stp_before_failover_part2.png, 17_stp_before_failover_part3.png, 18_stp_after_failover_part1.png
NAT-01	Test permitted Internet access	PAT translates internal traffic where ISP simulation supports it	NOT VERIFIED - Cloud-PT supplied no usable WAN address/next hop

6. Evidence register

Screenshots are stored under `evidence/` and indexed in [evidence/README.md](#).

- ✓ Final Packet Tracer topology and saved M2 file - 00_final_topology.png
- ✓ VLSM subnet and gateway verification - 01_engineering_dhcp.png
- ✓ All seven VLSM SVIs, masks, DHCP helpers, and ACL attachments - 26_dls1_vlsm_svis_and_helpers.png
- ✓ DHCP leases for representative VLANs - 01_engineering_dhcp.png , 02_admin_dhcp.png
- ✓ Guest and Finance DHCP leases - 08_guest_dhcp.png , 09_finance_dhcp.png
- ✓ ALS-ADM VLAN and trunk verification - 25_als_adm_vlan_and_trunk.png
- ✓ show vlan brief - 08_dls1_vlan_brief.png
- ✓ show interfaces trunk - 09_dls1_trunks.png
- ✓ show ip interface brief - 10_dls1_ip_brief.png
- ✓ show ip route and OSPF neighbour state - 11_dls1_ospf.png , 12_dls1_route.png
- ✓ Successful authorised server access - 03_engineering_server_ping.png
- ✓ Failed unauthorised server access - 04_production_server_ping_denied.png
- ✓ ACL hit counters after negative tests - 13_dls1_acls.png

- ✓ STP root and failover evidence - 15_stp_before_failover_part1.png , 16_stp_before_failover_part2.png , 17_stp_before_failover_part3.png , 18_stp_after_failover_part1.png
- ✓ Port-security evidence - 19_dls1_port_security.png , 20_port_security_violation_detected.png , 21_port_security_violation_status.png
- ✓ DHCP, DNS, and HTTP service screens - 05_svr_dc_dhcp.png , 06_svr_dc_dns.png , 07_svr_fp_http.png

7. Security policy: final ACLs

The seven named inbound ACLs below implement the CR4 access policy. The entries are listed in final IOS sequence order.

ACL	Final entries	Effect
ACL-ENG-IN	10 deny ip any 192.168.35.96 0.0.0.15; 20 permit ip any host 192.168.35.119; 30 permit ip any any	Engineering cannot reach Finance and can reach the file server
ACL-ADMIN-IN	10 deny ip any 192.168.35.96 0.0.0.15; 20 permit ip any host 192.168.35.119; 30 permit ip any any	Admin cannot reach Finance and can reach the file server
ACL-FIN-IN	10 deny ip any 192.168.35.0 0.0.0.255; 20 permit ip any host 192.168.35.119; 30 permit ip any any	Finance internal-access policy
ACL-PRD-IN	10 deny ip any host 192.168.35.119; 20 deny ip any 192.168.35.96 0.0.0.15; 30 permit ip any any	Production is denied the file server and Finance
ACL-ICT-IN	10 deny ip any 192.168.35.96 0.0.0.15; 20 permit ip any 192.168.35.112 0.0.0.15; 30 permit ip any any	ICT cannot reach Finance and may reach management/server VLAN 99
ACL-MGMT-IN	10 permit ip 192.168.35.64 0.0.0.15 any; 20 permit ip 192.168.35.112 0.0.0.15 any; 30 deny ip any any	Only ICT and management/server traffic is permitted
ACL-GUEST-IN	10 deny ip any 192.168.35.0 0.0.0.255; 20 permit ip any any	Guest traffic is denied internal project networks

The ACL definitions and counters are shown in evidence/M2_config_screenshots/13_dls1_acls.png .

8. Troubleshooting log

Date	Problem	Cause	Resolution	Evidence
2026-10-01	Engineering clients received APIPA addresses	Incorrect DHCP pool/gateway values and stale client lease state	Corrected the Engineering scope to 192.168.35.0/27 with gateway 192.168.35.1, then renewed DHCP	01_engineering_dhcp.png

2026-10-01	SVR-DC could not reach the routed network	VLAN 99 ACL did not permit the management/server subnet	Corrected the management ACL policy and verified reachability	13_dls1_acls.png, 12_dls1_route.png
2026-10-01	Trunk and cable labels did not match the worksheet	Physical Packet Tracer cabling used different switch-port labels	Updated the worksheet to match the actual ports and retained the redundant Engineering uplinks for STP testing	00_final_topology.png, 09_dls1_trunks.png
2026-10-01	DLS1 rejected trunk configuration	Catalyst 3560 required explicit 802.1Q trunk encapsulation	Added switchport trunk encapsulation dot1q before switchport mode trunk	09_dls1_trunks.png
2026-10-01	Native VLAN mismatch appeared on trunks	Worksheet labels did not match the physical cable mapping	Set the Engineering trunk native VLAN to 1 and Finance trunk native VLAN to 999	09_dls1_trunks.png, 09_als_fin_vlan_and_trunks.png
2026-10-01	ACL deny rules were ineffective	IOS sequence ordering placed permit rules before deny rules	Recreated the ACLs with deny entries first and verified hit counters	13_dls1_acls.png
2026-10-01	SVR-DC could not reach DLS1	Management ACL did not initially permit the server/management subnet	Corrected ACL-MGMT-IN and verified the routed path	12_dls1_route.png, 13_dls1_acls.png
2026-10-01	ALS-ADM uplink did not match the worksheet	The access switch used Gi0/1 to DLS1 Gi0/2, not the earlier assumed port pair	Corrected the worksheet and Gi0/2 trunk configuration	09_dls1_trunks.png

2026-10-01	Port-security violation required evidence	A test device/unknown MAC triggered the configured security policy	Captured the detected violation and resulting port status	20_port_security_violation_detected.png, 21_port_security_violation_status.png
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9. Final status

- R-EDGE Gi0/0 and Gi0/1: **Enabled and up**
- OSPF R-EDGE/DLS1 adjacency: **FULL**
- R-EDGE learned seven VLSM VLAN routes: **Verified**
- R-EDGE default route: **Configured in running configuration; Cloud-PT supplied no WAN address or next-hop, so the route is not installed as an active route**
- Packet Tracer implementation: **Complete and saved as** CLI066_Mbodi_40779750_M2_CONFIGURED.pkt
- Assigned VLSM feature: **Implemented and verified across the VLAN/SVI addressing plan**
- Services and ACLs: **Implemented and verified with DHCP, DNS, HTTP, authorised access, and denied access tests**
- Testing evidence: **Captured under** evidence/M2_config_screenshots/ **and** evidence/M2_service_tests/
- Evidence index: **Available at** evidence/README.md
- GitHub portfolio update: **Ready for publication**

10. Tested, passed, and not tested

Passed tests include VLSM addressing, representative DHCP leases, VLAN and trunk state, OSPF adjacency, internal routing, DNS and HTTP service availability, authorised Engineering server access, denied Production server access, ACL counters, STP failover/restoration, and port-security violation detection. The DHCP, DNS, and HTTP services were configured through Packet Tracer GUI screens, while routing, switching, STP, and ACLs were configured through IOS CLI.

NAT/PAT could not be fully tested because the Packet Tracer Cloud-PT handoff did not provide a usable WAN address or ISP next hop. This limitation is documented below rather than reported as a false pass.

11. Packet Tracer WAN limitation

The simulated Cloud-PT handoff brought R-EDGE GigabitEthernet0/0 to an up/up state but did not supply a DHCP address or ISP next-hop. Packet Tracer also rejected default-information originate always and ip route 0.0.0.0 0.0.0.0 dhcp on the selected IOS image. A supported interface-based default route and the OSPF default-information originate command were retained in the running configuration, while the absence of an active Internet default route is recorded rather than presented as a passing Internet test. Internal OSPF reachability to all seven VLSM VLANs was verified successfully.